

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_41c8dc8a-8a4\agent-a706b542f7cc32e86.jsonl`  
- SHA-256: `7a39a63c29ab85881e1a2f79ad3d7282b02c87cae0e2c94f213bd69dcc842350`  
- Source modified: `2026-07-02T03:00:07+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_41c8dc8a-8a4`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T02:58:05.142Z)

You are an adversarial verifier in a tech-debt audit. Your default stance: REFUTE. Today is 2026-07-02.

A finder claims the following about the repo at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer:

```
{  
  "title": "BACKLOG.md lists three shipped items as open, including a red 'blocks something'  
ByteTrack entry",  
  "category": "doc-rot",  
  "severity": "P2",  
  "file": "docs/BACKLOG.md",  
  "line": 39,  
  "evidence": "BACKLOG.md:39 marks 'supervision ByteTrack deprecation' as red/open, but the  
migration shipped in #386: backend/pipeline/detectors/person_detector.py:124 imports  
trackers.ByteTrackTracker and requirements.txt:29 pins 'trackers @ git+...@2.5.0'. BACKLOG.md:22  
lists 'App-factory / lifespan' as open, but create_app() exists at backend/main.py:93 (#370).  
BACKLOG.md:23 says 'orphaned Gemini uploads are never cleaned', but  
backend/providers/gemini.py:59 defines cleanup_orphaned_files() with a startup orphan sweep  
(#380). Doc header says 'Last updated: 2026-06-23'.",  
  "why_it_matters": "This stale state has already caused real confusion: the project memory  
feeding this very audit listed D13 (ByteTrack) as a remaining item. A red-flagged 'blocks  
something' entry that is actually done distorts session-start prioritization for every future  
session.",  
  "proposed_fix": "Docs-only PR: flip the three entries to done with PR links (#386, #370,  
#380), matching the checked-off convention already used at BACKLOG.md:24. Can ride in the same  
reconcile PR as the DOC_STATUS/README fixes.",  
  "effort": "S",  
  "tracked_in": ""  
}
```

Repo root candidates: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer,
C:/Users/avidu/Projects/khelsutra-guru/khelsutra, C:/Users/avidu/Projects/khelsutra-guru/sports-
data-collector, C:/Users/avidu/Projects/khelsutra-guru/rally-annotator,
C:/Users/avidu/Projects/khelsutra-guru/sports-obsreport (pick from the finding's repo field;
non-git dirs: 'C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup',
C:/Users/avidu/Projects/khelsutra-guru/khelsutra-screencast).

VERIFY against the CURRENT working tree (just pulled to origin master/main): (1) does the cited
code actually exist and behave as claimed at file:line? (2) was it already fixed by a recent
commit (git log --oneline -15 -- <file>)? (3) is the severity honest - construct the concrete
failure scenario or downgrade; (4) is the proposed fix sane and correctly scoped; (5) is it
owner-gated per: alpha/serving go-live, Studio P10/P11 corpus-promotion/training,
product/strategy/licensing, real cloud spend, counsel ToS/DPA?

HARD RULES: READ-ONLY (no file modifications, no installs, no test-suite runs). gh read-only
commands allowed. Your reasoning MUST cite file:line you actually read. CONFIRMED requires you

personally saw the defect/debt in current code; if you cannot reproduce the claim from the code, say REFUTED; if a commit fixed it, ALREADY_FIXED.

2. user (2026-07-02T02:58:11.673Z)

```
1      # Backlog ? deferred items to revisit
2
3      Single tracker for work we consciously deferred, so nothing gets lost between sessions.
Each item records what, why it's deferred, and the trigger to revisit. Every entry
traces to a real decision/PR/doc (no speculative work).
4
5      > Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all
of these to GitHub issues instead (or in addition).
6      > Status legend: ? blocks something · ? do when triggered · ? nice-to-have/cleanup.
7
8      _Last updated: 2026-06-23._
9
10     ---
11
12     ## Batch / declarative orchestration
13     - ? One-shot "serving startup-script" for the GPU VM ? `create_gpu_vm.sh` already
injects a metadata `startup-script` (`deploy/gcp/ops-agent-startup.sh`), but it only installs
the Ops Agent; standing up the live engine on the VM (`kheksutra/deploy/GPU_VM_SERVING.md`)
is still manual. Propose a `deploy/gcp/serving-startup.sh` fed via `--metadata-from-
file=startup-script=?` (or pasted into the GCP console) that, on boot: installs ffmpeg/cv2 libs
? pulls the engine source (from GCS) + builds the venv ? (optional) the WASB native env
(`gpu_setup.sh` + weights) ? writes the serving `config.json`/`engine.env` ? installs
`cloudflared` with the tunnel token ? starts `kheksutra-engine`. So a GPU serving box comes up
hands-off (reuses the `deploy/gcp/nightly_regression/run_on_vm.sh` bootstrap pattern). Why
deferred ? the design hinge is SECRETS: the Gemini key / CF-Access AUD / tunnel token must
not sit in a plaintext startup-script (instance metadata is broadly readable by anyone with
`instances.get`); they should come from Secret Manager (VM SA ? `secretAccessor`) or be
injected post-boot. Needs that small secrets-from-metadata design + which config is baked vs
fetched ? not just a script (owner asked 2026-06-23). Revisit: when the GPU-VM serving path
is used often enough to automate.
14     - ? One YAML manifest drives the whole bucket?reels run ? today, converting `gs://`
videos into highlight reels is a sequence of API calls (see
[`CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md`](CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md)). Propose a
declarative batch manifest (`pipeline.yaml`): inputs (bucket prefix/glob or explicit
list), compute target (local/cloud), detection (segmenter, skip_ai_handoff), per-video
selection rules (all / top-N longest / min duration / a query string), and output
(bucket prefix or local dir + watermark locale) ? driven by one `indexer --batch pipeline.yaml`
runner that loops process?query?select?compile?deliver, idempotent/resumable (skips already-
done `video_id`s), reusing the existing API + the compute-picker + the compute-path provenance.
Why deferred: needs a small design ? the manifest schema, the selection grammar,
idempotency/resume, and partial-failure handling ? i.e. a tracked project doc per
`PROJECT_DOC_TEMPLATE.md`, not a quick add (owner agreed 2026-06-23: ship the runbook doc now,
backlog the YAML driver). Revisit: when hands-off batch conversion is wanted ? start with a
design doc.
15
16     ## Owned-model / multisport (source: `OWNED_MODEL_IMPLEMENTATION_PLAN.md`,
`NEXT_STEPS.md`)
17     - ? Rename the repo off "badminton" ? owner-emphatic "ASAP" action; the project is
multisport. Touches remote name + README/CLAUDE.md framing + any hardcoded URLs. Revisit:
owner action, before broad external sharing.
18     - ? Quantify cost-to-Gen-2 (the largest open risk) ? benchmark real VRAM/token/$ for
the VLM SFT+GRPO on a rented A100/L40S (ms-swift multimodal-GRPO examples use 6?8 GPUs)
before standing up any owned serving stack. Revisit: before Phase-2 commitment.
19     - ? Define the ship-gate F1@tIoU target ? "very high P/R" must be operationalized
(floor = beat heuristic LOVO 0.480; learned is now 0.626 post metric-fix). Revisit: at M0.4
harness.
20
21     ## Engineering hardening deferred from the 2026-06-10 audit sweep
22     - ? App-factory / lifespan + lazy torch-cv2 imports ? `uvicorn backend.main:app`
leaves globals None, and importing the API pulls the GPU stack. Belongs with the async-job slice
```

```

(#16). **Revisit:** when the async slice starts.
23     - ? **WSL temp/frame/cache + Gemini remote-file GC** ? staged videos, PNG frame dumps,
`*_wasbcache`, and orphaned Gemini uploads are never cleaned. Needs a retention policy.
**Revisit:** before any volume/batch processing.
24     - ? **shlex-quote WSL config values** ? **DONE 2026-06-11 (#97):**
`detectors/base.py::wsl_tilde_quote` quotes everything after a leading `~/`, so quoting and
tilde expansion coexist; applied to every `~`-capable arg in both runners;
`tests/test_wsl_tilde_quoting.py` pins the command shapes (no quoted tilde can ever reach
`_wsl_bash`). Found live: the naive quoting had silently broken wasb_hybrid staging for the
default `~/clips`. Residual for tenancy-era: quote the remaining trusted-config interpolations
(repo_dir/conda_env) if config ever becomes per-tenant.
25
26     ---
27
28     ## Monetization / licensing
29     *(source: `archives/MONETIZATION_AUDIT.md` §7, ADR-005)*
30
31     - ? **Ultralytics Enterprise pricing** ? get a quote **only if** the
torchvision/ByteTrack detector (PR #8) proves insufficient on real footage. Currently *not
needed* (swap done). Quote-only pricing. **Revisit:** if detector quality forces reconsidering
YOLOv8.
32     - ? **Codec patent thresholds (AVC/HEVC/AAC)** ? confirm exact Via-LA (AVC) / Access
Advance (HEVC) royalty rates & free thresholds **with legal counsel** before relying on the
"small SaaS is under threshold" reading. Mitigation already chosen: emit royalty-free AV1/Opus
output. **Revisit:** before commercial launch.
33     - ? **Per-checkpoint weight/dataset licenses** ? if we borrow pretrained weights (YOLOX,
ActionFormer, WASB, TrackNetV3), the *code* license ? the distributed *weights/dataset* terms.
Either train our own or confirm each checkpoint. **Revisit:** when adopting any borrowed
weights.
34     - ? **Real GPU $/video & latency** ? benchmark the three LLM-step options (offline
ActionFormer vs self-hosted Qwen2.5-VL-7B vs Gemini paid-tier) on the target VM before pricing
the product. **Revisit:** when choosing the production LLM step.
35
36     ## Detector swap follow-ups
37     *(source: PR #8, ADR-005)*
38
39     - ? **`supervision` ByteTrack deprecation** ? `sv.ByteTrack` is deprecated as of
supervision 0.28.0 and **removed in 0.30.0** ; `requirements.txt` is pinned `<0.30.0` as a
stopgap. Migrate to `ByteTrackTracker` from the separate `trackers` package before lifting the
cap. **Revisit:** before bumping supervision past 0.30.
40     - ? **Real-video validation of the swap** ? the torchvision + ByteTrack Stage-1 is unit-
tested with mocks but **not yet validated on actual footage** for detection/tracking quality.
(ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video, so
expect tuning.) **Revisit:** when a real test clip + the pipeline are run together.
41     - ? **`yolo_hybrid` naming** ? the registry key, `yolo_` config keys, and `yolo-
hybrid-` DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining.
Optional rename. **Revisit:** if/when a clean break is worth the migration.
42
43     ## Golden-set subproject
44     *(source: `docs/GOLDEN_SET_IMPLEMENTATION.md`, `docs/GOLDEN_SET_VENDORING.md`)*
45
46     - ? **Annotation guideline** ? **DONE (2026-06-02):** authored as the Roora brief +
illustrated guide in the collector repo `docs/annotation/` ? rally CSV schema; ignore dead time
/ include every rally (incl. faults & lets) / shot = one contact; `ending_reason`
(forced/unforced-error taxonomy, shared across net-separated sports); singles & doubles for
badminton & pickleball.
47     - ? **Internal gold key** ? hand-annotate 5?10 licensed clips to near-perfect as the
vendor answer key. **Unblocked.** **Revisit:** anytime.
48     - ? **GS-3 ? `HumanVendorProvider`** ? concrete adapter for the chosen annotation vendor
(submit/poll/fetch + DPA/secure-link controls). **Vendor selected for the initial pilot: Roora**
(Bengaluru; iMerit / Taskmonk / TELUS remain the fallback shortlist). **Revisit:** once the
Roora pilot confirms quality, then build the adapter.
49     - ? **GS-4 oracle model** ? `AutomatedProvider` interface is stubbed (this work) with a
deterministic fake. Wire a *concrete* fast-tier auto-labeler later. **Oracle rule:** it MUST be
a different model lineage than production, or the regression test is blind to shared failures.

```

****Revisit:**** when a production LLM step is chosen (pick the oracle's lineage to differ).

50 - ? ****GS-5 ? regression triggers**** ? wire `regress()` into CI (pre-merge gate), on-demand CLI (the "surprise" case), and a scheduled run; include a deliberate re-baseline path.

****Revisit:**** after GS-4.

51 - ? ****Vendor pilot (Roora)**** ? run the paid pilot with ****Roora**** as a ****single-vendor pilot first**** (not a multi-vendor bake-off): assess quality-per-rupee against the \$4 bar; escalate to the iMerit/Taskmonk fallback only if it misses. ****Revisit:**** the pilot doubles as calibration once the gold key exists.

52 - ? ****Phase-1 video collection**** ? gather the round-1 golden corpus per `docs/GOLDEN\_SET\_PHASE1\_VIDEOS.md`: ****static self-recorded**** footage (mirror deployment, not broadcast), 3 sports incl. singles & doubles, across the diversity matrix; ~12 clips/sport with ****? held out by match****. ****Unblocked.****

53 - ? ****Train/eval separation guardrails**** (was `DEFERRED\_HARDENING\_PLAN` #13; design preserved here on archival 2026-06-14) ? train & eval videos live in ****separate catalogs**** today, kept apart by convention only; nothing ***enforces*** that a video used to train the learned segmenter (`backend/eval/training.py::build\_training\_set` ? `distill\_local`) is absent from the eval/regression set, so leakage would silently inflate the headline metrics. ****Concrete design**** (from the archived DEFERRED plan \$2): split unit = ****match, never clip**** (hold ~? of each sport as eval, per `GOLDEN\_SET\_PHASE1\_VIDEOS.md`); add a `split: "train"|"eval"` field to the collector's `curate export` manifest (cross-repo contract, ADR-002); ****[DONE ? pure-logic core, PR #171]**** `backend/eval/splits.py` now exists with `load\_split(manifest) -> (train\_ids, eval\_ids)` keyed by ****match id**** (not file MD5 ? re-encodes share a match but differ in `compute\_video\_id`) + `assert\_disjoint(train\_ids, eval\_ids)` that raises on overlap, with `tests/test\_splits.py` (overlapping ids ? `assert\_disjoint` raises). ****STILL DEFERRED:**** wire the guard into the training entrypoint (`distill\_local`/`training`) and into `run\_regression.py` (baseline computed strictly on the eval split); add the collector's `curate export` manifest `split` field (above); the integration test (a deliberately-leaked manifest fails the training entrypoint). Full rationale/risks: archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md \$2.

****Revisit:**** dedicated design session with the user (needs a collector-side manifest change).

54

55 **## Golden regression fixtures (video-free, non-attributable regression suite)**

56 ***(source: `docs/GOLDEN\_REGRESSION\_FIXTURES.md` \$7 tracker / \$9 definition-of-done)***

57

58 Core suite ****shipped**** (M1.M2.M3.MX.P1.P2.P3.DOC ? ? synthetic + 6 de-attributed real `fx\_r01`?`fx\_r06` fixtures replay cross-OS, leak-guarded, CONTROL-only). The two items below are the ****gating remainder**** of the doc's \$9 definition-of-done (it flips to `DONE`/archive only when ****P0 + M1?M4 + P1 + DOC**** are ?); 01/02 are explicitly optional. Tracked here so the track isn't silently treated as "done" while DoD items remain.

59

60 - ? ****M4 ? quality-floor gate + first `regression\_baseline.json`**** ***(recommended-next; unblocked).*** Assert per-fixture ****and**** mean F1 ? a committed floor (`tests/test_golden_quality_floor.py` + `eval_baselines/regression_baseline.json``). ***First move:*** seed the per-slug floor from the F1 already frozen in each `fx\_\*/expected.json` `score` (real fixtures sit at ****0.0?0.59****, so floor = `current ? tolerance`, not a global bar). ****Gotchas:**** record a ****windowing fingerprint**** in the baseline (else a SERVED retune silently passes); keep it ****CONTROL-only**** (learned/numpy-BLAS is non-deterministic across CPUs); add it to the `golden-crossos` CI matrix. ****Revisit:**** anytime ? it's the cheapest step to flip the track to DONE.

61 - ? ****P0 remainder ? functional video-name rename**** ***(deferred per owner safe-scope, 2026-06-16).*** Rename the functional names (`eval\_baselines/\*.json` keys, `rally\_seg\_eval.MULTICOURT\_CLIPS`, quality-log names) ? opaque ids. ***Why it's hard:*** the keys feed `training/gen0/harness.py`'s LOVO floor matched to the ****gitignored `output/human\_lovo\_manifest.json` + data files**** ? rename them together or the floor breaks; also collides with the hot quality track. ***First move:*** put the names in a gitignored `.leakscan\_private` and run `python -m tools.leak_scan --names .leakscan_private`` to enumerate every occurrence, then plan the coordinated rename. ****Revisit:**** before any open-sourcing / public commit.

62 - ? ****01 ? protobuf + deterministic serialization**** ***(optional).*** `rally.fixture.v1` proto3 + codegen + `protoc`/`git diff --exit-code` guard. ****Revisit:**** only if cross-OS ****byte-stable**** snapshots are wanted (JSON parse-compare suffices today).

63 - ? ****02 ? Tier-1 key-gated path**** ***(optional).*** `scripts/fetch\_golden.py` (OneDrive token) and/or SOPS+age, `RALLY\_GOLDEN\_DIR` wired, Tier-1 CI gated on a secret (never on fork runners). ****Revisit:**** only if sharing person/biometric streams (Tier-1) becomes necessary.

64

```

65     ## Recording best practices
66     *(source: `docs/VIDEO_RECORDING_GUIDELINES.md`)*
67
68     - ? **Validate recording hypotheses** ? empirically test which capture conditions
        (resolution, fps, camera angle/height, lighting) actually move precision/recall, then promote
        the winners into user-facing "how to record" guidance. Only the static-camera rule is currently
        evidence-backed (ADR-004). **Revisit:** once a controlled corpus + the eval loop can measure
        per-condition deltas.
69
70     ## Code & test organization
71     *(source: owner review comments on PR #35, 2026-06-07)*
72
73     - ? **`tests/` reorganization to reduce fragmentation** ? owner wants to isolate test
        components on a grouping
74         metric (by subsystem/component) so `tests/` is less fragmented, mirroring the new
75         `backend/pipeline/audio/` +
76         `backend/eval/audio/` layout (e.g. audio tests grouped under `tests/audio/`, eval
        tests under `tests/eval/`).
77         The exact grouping **metric is the owner's design choice** (consolidate small files
        vs. add per-area dirs ? the
78         "lesser fragmentation" goal slightly tensions with adding dirs), so this is left as a
        **dedicated refactor PR**,
79         not guessed here. **Revisit:** when the owner specifies the grouping scheme. (The
        `backend/eval/audio/` half of
80         the request ? comment 1 ? was done in PR #35.)
81
82     ## Commercial Readiness & Low-Regret Moves ? Approved Decisions Checkpoint (2026-06-07)
83     *(source: Commercial Readiness Review + structure/sequencing discussion with Grok as
        second-in-command)*
84
85     - ? **Docs-only checkpoint PR** ? This update + the refreshed
        `docs/archives/past_projects/low-regret-moves/LOW_REGRET_MOVES_PLAN.md` (since archived)
        formally record the approved path before any code changes. The initial review docs (PR #37)
        introduced the findings; this entry locks the execution decisions.
86     - ? **Approved directory structure** ? Adopt a layered approach under `backend/` for
        professionalism and future growth (no flat dumping of new modules, no explosion of 15+ single-
        file dirs):
87         - `backend/config/` (package ? start with this)
88         - `backend/api/` (FastAPI app, routes, static UI)
89         - `backend/pipeline/` (validators + segmenters + stitcher + audio + detectors)
90         - `backend/infrastructure/` (database + external adapters)
91         - Keep `backend/eval/` prominent as a first-class concern.
92         - Logical grouping for providers, sports, annotations, utils, and a small `core/` only
        when needed.
93     - ? **Sequencing (Config first, then atomic refactor)** ? Implement Deliverable 1
        (Config) **first**, introducing the `backend/config/` package immediately. Perform the full
        directory structure refactor as a **separate high-pri atomic PR** afterwards. This minimizes
        conflicts, keeps PRs focused, and gives Claude a clean window with no concurrent moves.
94     - ? **Deliverable 1 (Config)** ? `backend/config/` package (`models.py` + `loader.py`)
        instead of a single `config.py`. Typed `AppConfig`, central loader with built-in defaults + file
        override, `extra="allow"` for migration. Update `setup.py` + `main.py`. Add tests. Land this
        before the layout move.
95     - ? **Subsequent items** ? See the refreshed `docs/archives/past_projects/low-regret-
        moves/LOW_REGRET_MOVES_PLAN.md` (archived) for the full ordered list (Error contracts, logging,
        packaging, detector abstraction, etc.). The structure refactor is now explicitly tracked as its
        own high-pri step right after Config.
96
97     **Status legend reminder:** ? = approved / in progress tracking, ? = do when triggered.
98
99     ## Observability reports ? shipped; intersections with the low-regret items (2026-06-07)
100    *(source: per-video observability feature ? indexer PR #41/#42/#43, collector PR
        #11/#12; shared lib `obsreport`)*
101
102    - ? **Low-regret items 1 (Config pkg, PR #39) + 2 (directory restructure, PR #40) are

```

DONE.** Remaining = items 3?9 in `docs/archives/past\_projects/low-regret-moves/LOW\_REGRET\_MOVES\_PLAN.md` (archived).

103 - ? **Per-video observability reports shipped in BOTH tools** (a TECHNICAL report + a CUSTOMER-friendly summary), via the **soft-dependency** lib `obsreport` (github.com/avidullu/sports-obsreport, MIT, v0.1.3). Each run writes `

104 - ? **Item 8 (Input-Quality UX) is LARGELY SUBSUMED** by these reports (validation stage + footgun flags + verdict already surface quality; a clean customer summary already exists). **Do NOT build a parallel mechanism** ? instead EXTEND the report: add an audio-readiness stage/flag to `backend/reporting/` (harvest) + the obsreport spec. **Revisit:** when item 8 is picked up.

105 - ? **Item 7 (unify hybrid segmenters) should land before the report's real AI-provider timing.** The report records AI timing as `not\_captured`; once `wasb\_hybrid`/`tracknet\_hybrid` are unified there is ONE place to thread a recorder callback through `provider.analyze\_video` ? the report. **Revisit:** with item 7 (Claude to add the timing hook there).

106 - ? **Item 3 (Error/Result contract) synergizes with the report.** Today the report INFERS failures (e.g. `silent\_provider\_noop` = 0 segments + no API key). A structured Result/error from providers/segmenters would let the report read the real reason as a first-class flag. **Revisit:** with item 3.

107 - ? **`obsreport` is a SOFT/commented dep** in `requirements.txt` (works locally via editable install; pipeline no-ops if absent). **Pin the `git+ssh@v0.1.3` tag** in both repos once SSH/CI access to the private lib repo exists.

108 - ? **`run\_all\_tests.py`** added at repo root ? run the full suite (clear PASS/FAIL, pytest exit code) before sending PRs.

109 - ? **Config-migration follow-ups (DONE ? config & identity consistency pass)** ? the two typed-config bugs from

110 #39/#40 are fixed: (1) **`/api/compile` now honors `--output-dir`** and (2) **`output\_dir` is a real field on

111 `OutputConfig`** (default "output"); the CLI overrides it on the typed model in `main()`, retiring the

112 write-only `\_runtime\_overrides` hack. Same pass also unified `video\_id` MD5 derivation into

113 `backend/utils/hashing.py::compute\_video\_id` and migrated all four `run\_\*.py` scripts off raw `json.load` onto

114 `backend.config.load\_config`. See `CODE\_MAP.md` §2.0/§6 (footguns retired).

115

3. user (2026-07-02T02:58:13.785Z)

b3c6c2b docs(backlog): one-shot serving startup-script for the GPU VM (#347)

2f4ecad docs(runbook): GCS-bucket videos ? highlights on the L4 Cloud Run + committed build config (#346)

9b37269 docs(backlog): track golden-regression-fixtures DoD remainder (M4, P0 rename) + optional 01/02 (#323)

9736a3b docs: re-ground stale doc claims to the current code (multi-agent audit) (#242)

e69e23d docs: fix doc drift across 16 docs (#173)

1ca57c3 docs: archive DEFERRED_HARDENING_PLAN ? all items dispatched (#16 done, #7 done via #167, #13 ? BACKLOG)

020ff61 docs: status sync - PR-2 MERGED (#99, live-E2E-verified), sweep recorded, BACKLOG shlex item closed (#100)

62fb2fb docs: sync to ratified decisions + repo reality (audit truth-sync) (#71)

3fd7232 docs: handoff-consistency sweep + strategy checkpoint + CLAUDE.md (#59)

d965845 docs: retire footguns fixed by this change (CODE_MAP + BACKLOG)

f9b8313 docs: fix staleness in roadmap and audio/decision docs

5b39068 docs+tooling: regenerate CODE_MAP for layered layout + run_all_tests.py + Grok low-regret notes

9029415 docs: checkpoint approved structure + sequencing decisions (Config first, atomic refactor follow-up)

8718f1f docs: add Commercial Readiness Review + Low-Regret Moves starter plan (Config as Deliverable 1) (#37)

64ccc50 feat(audio): shuttle-hit detector + E1 go/no-go probe (ADR-007) (#35)

e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)

81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)

8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)

4. user (2026-07-02T02:58:19.808Z)

```
100         # weights="DEFAULT" pulls the best available COCO-pretrained checkpoint.
101         self.model = builder(weights="DEFAULT")
102         self.model.eval()
103         self.model.to(self.device)
104
105     def make_tracker(self, fps: float) -> Any:
106         """Create a fresh ByteTrack tracker for one chunk (track IDs are chunk-local).
107
108         D13/P12: migrated from ``supervision.ByteTrack`` (deprecated, removed in 0.30.0)
109         to the standalone ``trackers`` package (``ByteTrackTracker``). Imported lazily
110         so
111         tests can mock this method without the package installed.
112
113         The constructor params are pinned to ``sv.ByteTrack``'s behaviour so the
114         migration
115         is behaviour-preserving. The new package's defaults differ materially ?
116         ``track_activation_threshold`` 0.7 (vs sv's ~0.35 effective floor),
117         ``high_conf_det_threshold`` 0.6 (vs sv's ~0.35), and
118         ``minimum_consecutive_frames`` 2 (vs sv's 1). Left at the new defaults, a person
119         detected at confidence [0.5, 0.7) (common during fast smashes / partial
120         occlusion)
121         would never spawn a track, silently regressing rally recall; and the first
122         sampled
123         frame of every track would be emitted as -1 (dropped). Pinning all three keeps
124         the
125         0.5 ``detector_score_threshold`` pre-filter the effective floor and yields an ID
126         on
127         the first matched frame. ``minimum_iou_threshold`` is left at the package
128         default
129         (0.1), which is roughly equivalent to sv's matching gate.
130         """
131         from trackers import ByteTrackTracker
132
133         frame_rate = max(1, int(round(fps)))
134         return ByteTrackTracker(
135             frame_rate=frame_rate,
136             track_activation_threshold=0.25, # sv default; <= detector_score_threshold
137             (0.5)
138             high_conf_det_threshold=0.35, # match sv's det_thresh so [0.5,0.7) can
139             spawn
140             minimum_consecutive_frames=1, # sv default: emit a real ID on the first
141             frame
142         )
143
144     def detect_and_track(
145         self, frame: Any, tracker: Any, score_thresh: float
146     ) -> List[Tuple[int, Tuple[float, float, float, float]]]:
147         """Detect persons in a frame and assign persistent track IDs.
148
149         Returns a list of (track_id, bbox) where bbox is (x1, y1, x2, y2) in pixels.
150         This replaces the old ultralytics ``model.track(...)`` call, which bundled
151         detection AND association: torchvision does the detection, supervision's
152         ByteTrack does the association.
153         """
154         import supervision as sv
155         import torch
156
157         # cv2 frames are BGR uint8 HxWxC; torchvision wants RGB float [0,1] CxHxW.
158         rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
159         tensor = (
```

5. user (2026-07-02T02:58:20.720Z)

```

18:supervision>=0.21.0
19:# D13/P12: ByteTrack moved out of `supervision` (removed in 0.30.0) to the standalone
20:# `trackers` package. The PyPI wheel is BROKEN (ships metadata only, no source ? see
21:# https://github.com/roboflow/trackers/issues/474), so install from git until fixed:
22:# pip install "trackers @ git+https://github.com/roboflow/trackers.git@2.5.0"
23:# (Apache-2.0; depends on supervision >=0.26.1, so it composes cleanly.)
24:# NOTE: roboflow/trackers uses bare version tags (2.5.0), NOT v-prefixed.
29:trackers @ git+https://github.com/roboflow/trackers.git@2.5.0

```

6. user (2026-07-02T02:58:30.901Z)

```

93:def create_app(config: AppConfig, db: Database) -> FastAPI:

```

7. user (2026-07-02T02:58:31.672Z)

```

25:# 01/P11: every upload is tagged with this display_name prefix so the startup orphan
28:# are ever deleted by cleanup_orphaned_files().
59:     def cleanup_orphaned_files(self, grace_sec: int = 3600) -> int:
60:         """Sweep and delete video files orphaned on the Gemini File API by a previous
107:             "[Gemini GC] deleted %d orphaned video file(s) older than %ds.",

```

8. user (2026-07-02T02:58:46.427Z)

```

90:app = FastAPI(title="Sports Highlight Indexer API")
93:def create_app(config: AppConfig, db: Database) -> FastAPI:
2184:     create_app(global_config, db_instance)

```

9. user (2026-07-02T02:58:47.700Z)

```

3624a8d feat(D13/P12): migrate ByteTrack to standalone trackers package (#386)
cd5b3d2 feat(01/P11): Gemini remote-file GC ? startup orphan sweep (#380)
de09ec8 refactor(main): app-factory + typed-config migration (D2/D3/P2/P3/P5/P7) (#370)

```

10. user (2026-07-02T02:59:02.407Z)

```

1     import argparse
2     import csv
3     import json
4     import logging
5     import math
6     import os
7     import shutil
8     import socket
9     import sys
10    import tempfile
11    import threading
12    import uuid
13    import webbrowser
14    from importlib.metadata import PackageNotFoundError
15    from importlib.metadata import version as _pkg_version
16    from urllib.parse import quote
17
18    import uvicorn
19    from dotenv import load_dotenv
20
21    # PACKAGING_PLAN.md P0a: prefer an app-home .env ONLY when RALLY_APP_HOME is set (the
packaged
22    # launcher pins it); otherwise env_file_to_load() returns None and load_dotenv(None)
keeps the
23    # exact historical frame-relative walk-up ? byte-identical to the pre-P0a bare
load_dotenv().
24    # stdlib-only import ? keeps this pre-pydtantic line light.
25    from backend.utils.app_home import env_file_to_load, resolve_output_dir
26

```



```

27     load_dotenv(env_file_to_load())
28
29     # Centralized logging setup for the indexer (hot paths use logger, CLI output uses print
for direct user-facing messages).
30     logging.basicConfig(
31         level=logging.INFO,
32         format="%(asctime)s [%(levelname)s] %(name)s: %(message)s",
33         datefmt="%H:%M:%S",
34     )
35     logger = logging.getLogger(__name__)
36     from typing import Any, Dict, List, Optional, Tuple
37
38     from fastapi import FastAPI, HTTPException, Request
39     from fastapi.middleware.cors import CORSMiddleware
40     from fastapi.responses import FileResponse, HTMLResponse, JSONResponse
41     from fastapi.staticfiles import StaticFiles
42     from pydantic import BaseModel
43     from starlette.middleware.trustedhost import TrustedHostMiddleware
44
45     from backend import (
46         billing, # M8: per-job cost ledger (USD internal / INR display)
47         storage, # M2: URI-aware input acquisition (local = identity passthrough)
48     )
49     from backend.api import (
50         auth as edge_auth, # M9: Cloudflare Access edge-auth (default-OFF)
51     )
52     from backend.config import (
53         AppConfig,
54         StartupCheckError,
55         StitchingConfig,
56         enforce_startup_checks,
57         write_light_default_config, # P2b: batteries-included first-run config seeding
58     )
59     from backend.config import ( # new typed config
60         load_config as load_app_config,
61     )
62     from backend.infrastructure.database import Database
63     from backend.pipeline.segmenters.base import segmenter_registry
64     from backend.pipeline.stitcher.compiler import VideoCompiler
65     from backend.pipeline.stitcher.watermark_i18n import localize_watermark
66     from backend.pipeline.validators.base import registry
67     from backend.providers import provider_registry
68     from backend.reporting import emit_indexer_report
69     from backend.results import Failure # standardized error/result contract
70     from backend.storage.capabilities import (
71         storage_capabilities, # P3: secret-free "which media can this box use" report
72     )
73     from backend.utils.ffmpeg import ( # P2a: single ffmpeg/ffprobe resolver
74         ffmpeg_bin,
75         ffprobe_bin,
76     )
77     from backend.utils.freespace import require_free_bytes # HTTP-507 disk guard (P2, D11)
78     from backend.utils.hashing import compute_video_id # canonical file-identity hashing
79     from backend.utils.paths import (
80         output_base,
81         resolve_under_root,
82         safe_output_filename,
83         validate_input_path,
84     )
85     from backend.utils.redact import (
86         redact_secrets, # P0b: mask secret-shaped fields in /api/config
87     )
88     from backend.utils.single_instance import acquire_lock # P0e: one instance per database
89
90     app = FastAPI(title="Sports Highlight Indexer API")

```

```

91
92
93     def create_app(config: AppConfig, db: Database) -> FastAPI:
94         """Factory: inject shared state via ``app.state`` so route handlers access it
through
95         ``request.app.state`` ? eliminating the Optional module-level globals that forced
96         ``backend.main`` and ``backend.api.job_worker`` into mypy quarantine.
97
98         Also sets the module-level ``global_config`` / ``db_instance`` for transitional
99         compatibility (helpers and tests still reference them). Route handlers use
100         ``request.app.state`` exclusively."""
101         global global_config, db_instance
102         global_config = config
103         db_instance = db
104         app.state.config = config
105         app.state.db = db
106         return app
107
108
109     def _cors_origins_from_env(env: Optional[Dict[str, str]] = None) -> List[str]:
110         """M9: CORS allow-origins from the ``RALLY_CORS_ALLOW_ORIGINS`` env var (comma-
separated),
111         default ``["*"]``. Read from the ENV (not a cwd ``config.json``) so importing
``backend.main``
112         does zero filesystem I/O ? a hosted/multi-tenant deploy sets this to lock CORS to
its origin."""
113         src = os.environ if env is None else env
114         raw = (src.get("RALLY_CORS_ALLOW_ORIGINS", "") or "").strip()
115         origins = [o.strip() for o in raw.split(",") if o.strip()]
116         return origins or ["*"]
117
118
119     # CORS for the web UI. allow_origins='' with allow_credentials=True is rejected by
browsers per the
120     # fetch spec + is unsafe to carry into multi-tenant; this tool uses no cookies, so
credentials stay
121     # off. The origin list is configurable via the env var above (default ["*"]) so a hosted
deploy can
122     # lock it to its origin; the middleware is fixed at startup.
123     app.add_middleware(
124         CORSMiddleware,
125         allow_origins=_cors_origins_from_env(),
126         allow_credentials=False,
127         allow_methods=["*"],
128         allow_headers=["*"],
129     )
130

```

11. user (2026-07-02T02:59:19.259Z)

```

backend/pipeline/detectors/person_detector.py:34:import cv2
---
23:- ? **WSL temp/frame/cache + Gemini remote-file GC** ? staged videos, PNG frame dumps,
``*_wasbcache``, and orphaned Gemini uploads are never cleaned. Needs a retention policy.
**Revisit:** before any volume/batch processing.

```

12. user (2026-07-02T02:59:20.810Z)

```

e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)
aa915a7 deploy/gcp: centralize project + bucket names in one sourced buckets.env (#410)
52d97b5 P4: POST /api/assets ? store a video into the user's chosen medium (#409)
7ea6ea9 P3: /api/capabilities reports usable storage backends (#408)
f28af73 P2: engine free-space guard (HTTP 507 + WSL frame-stage BLOCK) (#407)

```

759a67f feat(api): frame-preserving annotation proxy for /api/source (D4) (#406)
master...origin/master

13. user (2026-07-02T03:00:06.987Z)

Structured output provided successfully